

Air conditioning (ME40107)

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Lecture 1: Introduction & Basic concepts

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Introduction to air conditioning

Some questions!

Q1. What is air conditioning?

Ans. Treating or conditioning of atmospheric air so as to control simultaneously its:

- **Temperature**
- **Moisture content**
- **Movement**, and
- **Purity**
- In addition to the above, it is also essential to control the **noise level** in the conditioned space
- → Sometimes these systems are also called as **Environment Control Systems** or ECS

Q2. Why do we need air conditioning?

- Ans.**
- 1. To provide thermal comfort to human beings**
 - 2. To cater to various industrial processes and products**

Q2.1. What is thermal comfort?

Ans.: “State of mind that expresses satisfaction with thermal environment”

Q2.2. How do we achieve thermal comfort?

- Ans.:** a) By conditioning the thermal environment to suit our needs, **or**
b) By adapting ourselves to suit the thermal environment

Q2.3. How do we condition the thermal environment?

Ans.: By controlling air and surrounding temperatures, moisture content, air motion, purity, noise level etc.

Q2.4. How do we adapt ourselves to the thermal environment?

Ans.: By changing our clothes, diet, activity, posture, habits etc.

Q3. How do we achieve air conditioning?

Ans. By choosing equipment which can perform the processes of:

1. **Cooling or heating to control air temperature**
 2. **Adding or removing water vapor** to control moisture content
 3. **Imparting motion** to control air movement, and
 4. **Filtration** to control air purity
- Suitable **noise/vibration attenuation** techniques are used to maintain the required noise levels.

Q4. How do we classify air conditioning systems?

Ans. Classification can be done in several ways

1. Depending on principle:

- Passive air conditioning
- Active air conditioning

2. Depending on application:

- Comfort/Industrial/Commercial

3. Depending on configuration:

- Central/Package/Room units

4. Depending on method of carrying heat:

- All water/All air/Air-water/Direct expansion

Q5. What are the benefits & costs of air conditioning?

- **Benefits:**

- Improvement in performance/productivity
- Decrease in absenteeism at workplace
- Reduction in rejections and better product quality
- Reduced drinking water consumption, time spent on make-up!

- **Costs:**

- High levels of energy consumption
- Adverse effect on environment (global warming)
- Sick building syndrome
- Possible long term effects on overall health?

A brief history of air conditioning

- The **desire for comfort** is a **basic instinct** of all living beings
- Hence **air conditioning**, that deals with thermal comfort is as old as human beings
- **Early attempts** by human beings for comfort led to:

1. **Discovery of fire**

2. **Use of clothing**

3. Seeking comfort in **caves**, followed by **construction of houses**

4. Invention of **passive (or natural) cooling techniques** such as:

1. Use of **natural ventilation**

2. Use of **evaporative cooling**

3. Use of **nocturnal cooling**

4. Use of **naturally available ice** etc.

Limitations of passive cooling

- **Effectiveness** of method used **varies** from **place to place** and **season to season**
- Can be **uncertain** and **unsteady** as it is **weather dependant**
- Providing **cooling on large scale** could be **difficult** and **expensive**
- Imposes **restrictions on building design**
- **Not much freedom for architects** in terms of **design**, use of materials etc
- **Not much scope** for **commercialization** and **business growth**

Advent of artificial air conditioning

- **Coincides** with **industrial revolution** and development of **heat engines**, **refrigeration systems** etc
- The **real impetus** was provided by:
 - development of synthetic working fluids (e.g. **CFCs**),
 - **electrical motors, instruments & controls**
- **Growth of cities & need for cheap and comfortable houses**, increasing **manufacturing & commercial activities** fuelled the growth

Artificial air conditioning

- **Artificial air conditioning** is developed using the **principles** of:
 - ① – Vapour compression refrigeration cycles driven by mechanical/electrical power
 - ② – Vapour absorption refrigeration cycles driven by **heat**
 - ③ – Thermoelectric systems driven by electrical power
- Of the above, air conditioning systems based on **vapour compression refrigeration cycles** driven by **mechanical/electrical power** are **most common**

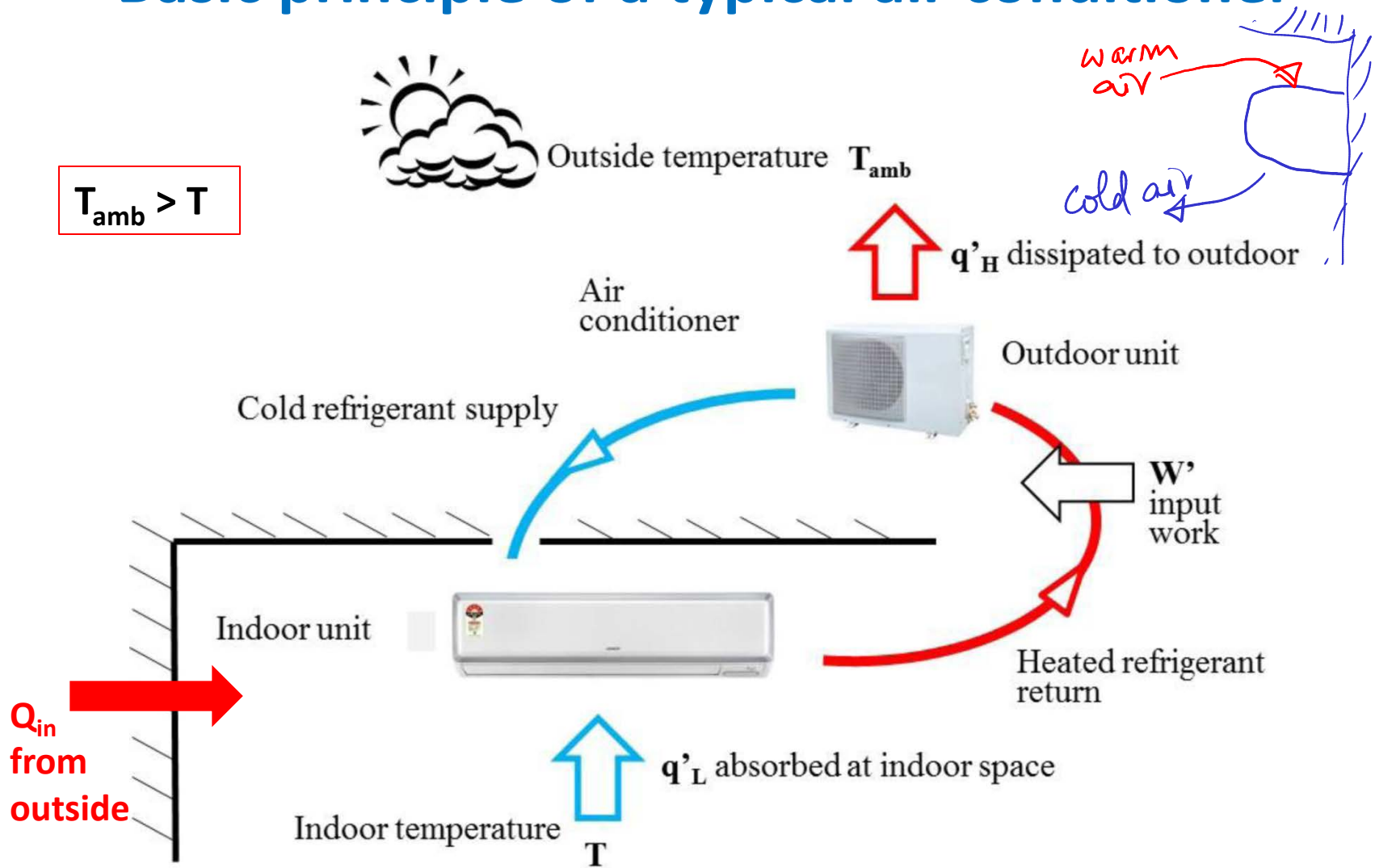
Characteristics of artificial air conditioning

- o Can **maintain required space conditions** irrespective of external climate, weather etc
- o Can be made extremely reliable and rugged
- o Can **maintain any condition** inside the conditioned space, which is **very important** for many **industries, special applications** etc.
- o Is **suitable** for **both small and large scale** applications
- o **Fewer restrictions** on **building type, materials used etc**
- o Provides lots of freedom to the architect
 - o Made the construction of **block type houses & sky scrapers** possible!
- o Is inexpensive to own with mass scale production
- o Provides opportunities for commercialization & business
- o Highly **energy intensive** with possibly **adverse environmental effects**

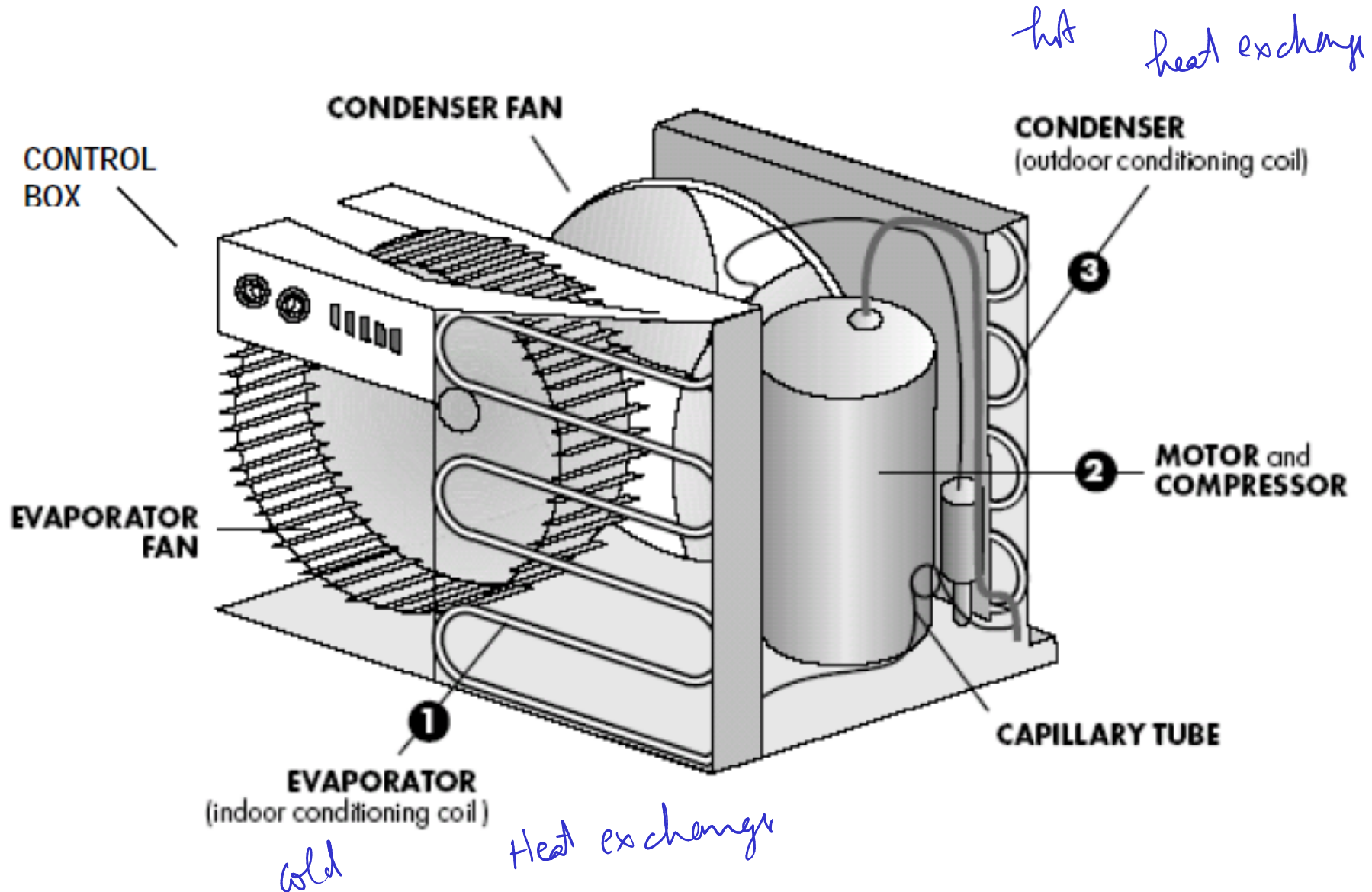
Impact of artificial air conditioning

- o Led to the **growth of densely populated cities** with skyscrapers
- o Made **conversion of the uninhabitable parts** of the world into **inhabitable parts**
- o Led to the **growth** of several **industries** such as **pharmaceuticals, healthcare, electronics, space travel, printing, chemicals** etc.
- o Hugely impacted **movement of people** through **transport air conditioning**
- o Provided **improved working conditions** at work places
- o Provided **almost unlimited freedom to the architects!**

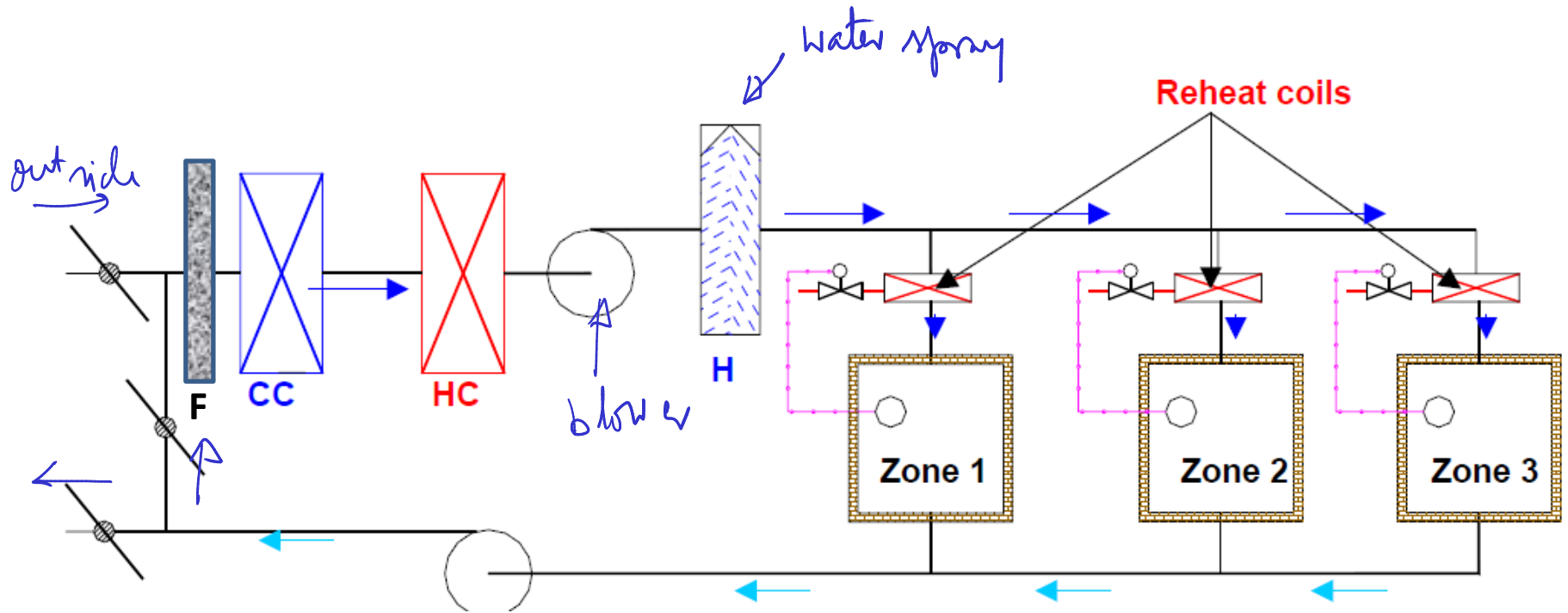
Basic principle of a typical air conditioner



A window air conditioner without cover!



A typical all year, air conditioning system



F: filter

CC: cooling coil

HC: heating "

Modern trends in air conditioning

- o Development of **efficient air conditioning and building management systems**
- o Use of **smart materials**, e.g. smart glass
- o Judicious mix of **passive** and **active cooling** techniques
- o Increased emphasis on **indoor air quality**
- o Use of **adaptive thermal comfort** standards
- o Use of **personalized cooling systems** with **personal controls**
- o Use of **renewable energy** for driving air conditioning systems
- o Growing concern for **outside thermal comfort!**

Course Objectives

At the end of the course, the student should be able to:

- 1) Clearly **understand** the meaning, need and impact of air conditioning
- 2) **Perform** simple psychrometric calculations
- 3) **Understand** the use of relevant thermal comfort standards
- 4) **Decide** conditions required for thermal comfort
- 5) **Perform** simple cooling load calculations
- 6) **Select** suitable air conditioning system for a given application
- 7) **Estimate** the size of air conditioning ducts
- 8) **Estimate** the power consumption due to air conditioning

End of Lecture 1